Chapter 4
Solution of Electrostatic Problems

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Contents

4.1 Introduction

- **Problems of electrostatic field: <u>electric potential, electric field intensity, and electric charge distribution</u>.**

- **In free space (boundless), if the charge distribution is given, the electric field intensity and the electric potential can be found by**

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\int_{V'} \frac{\rho(r - r')}{|r - r'|^3} dv', \quad \varphi = \frac{1}{4\pi\varepsilon_0}\int_{V'} \frac{\rho}{|r - r'|} dv'$$

- **In many practical problems the exact charge distribution is unknown (e.g., the charge distribution on a conducting boundary).**

- **Differential equations must be solved subject to the appropriate boundary conditions. These are *boundary-value problems*.**

- **When the conducting bodies have boundaries of a simple geometry, the *method of images* may be used.**

4.2 Poisson's and Laplace's Equation

- The two fundamental governing equations for electrostatics are

$$\begin{cases} \nabla \cdot \vec{D} = \rho_f \\ \nabla \times \vec{E} = 0 \end{cases}$$

- The electric potential is defined such that

$$\vec{E} = -\nabla \varphi$$

The differential equation of φ

In a simple medium:

linear and isotropic, ε

is a constant.

$$\left.\begin{aligned} \nabla\cdot\vec{E} &= \rho/\varepsilon_0 \\ \vec{E} &= -\nabla\varphi \end{aligned}\right\}$$

Scalar poisson equation

$\vec{\mathbf{E}}_1 = -\nabla\varphi_1$

Sub−1 ε_1 $\quad \nabla^2\varphi_1 = 0$

$\nabla^2\varphi = -\rho/\varepsilon$

Charge

Sub−2 ε_2 $\quad \nabla^2\varphi_2 = 0$

$\vec{\mathbf{E}}_2 = -\nabla\varphi_2$

$$\boxed{\nabla^2\varphi = -\rho/\varepsilon}$$

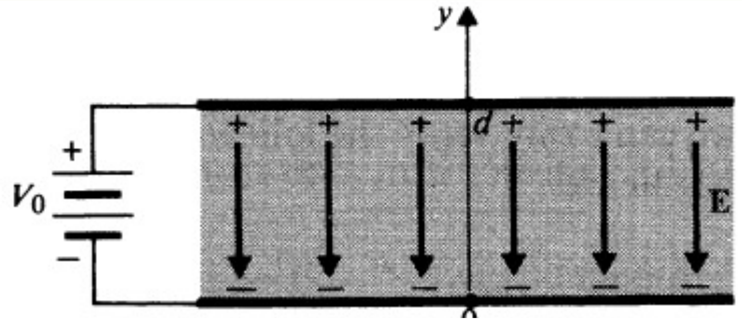
When, $\rho = 0$

$$\boxed{\nabla^2\varphi = 0}$$

Laplace equation

Cartesian Coordinates:

Ex. **The two plates of a parallel-plate capacitor are separated by a distance d and maintained at potentials 0 and V_0. Assuming negligible fringing effect at the edges, determine (a) the potential at any point between the plates, (b) the surface charge densities on the plates.**



Solution

a) Laplace's equation is the governing equation for the potential between the plates, since $\rho = 0$ there. Ignoring the fringing effect of the electric field is tantamount to assuming that the field distribution between the plates is the same as though the plates were infinitely large and that there is no variation of V in the x and z directions. Equation (4–7) then simplifies to

$$\frac{d^2V}{dy^2} = 0, \qquad (4\text{–}11)$$

where d^2/dy^2 is used instead of $\partial^2/\partial y^2$, since y is the only space variable here.

Integration of Eq. (4–11) with respect to y gives

$$\frac{dV}{dy} = C_1,$$

where the constant of integration C_1 is yet to be determined. Integrating again, we obtain

$$V = C_1 y + C_2. \tag{4-12}$$

Two boundary conditions are required for the determination of the two constants of integration, C_1 and C_2:

$$\text{At } y = 0, \qquad V = 0. \tag{4-13a}$$

$$\text{At } y = d, \qquad V = V_0. \tag{4-13b}$$

Substitution of Eqs. (4–13a) and (4–13b) in Eq. (4–12) yields immediately $C_1 = V_0/d$ and $C_2 = 0$. Hence the potential at any point y between the plates is, from Eq. (4–12),

$$V = \frac{V_0}{d} y. \tag{4-14}$$

- **The potential increases linearly from y=0 to y=d**

b) In order to find the surface charge densities, we must first find $\mathbf{E}$ at the conducting plates at $y = 0$ and $y = d$. From Eqs. (4–3) and (4–14) we have

$$\mathbf{E} = -\mathbf{a}_y \frac{dV}{dy} = -\mathbf{a}_y \frac{V_0}{d}, \tag{4–15}$$

which is a constant and is independent of y. Note that the direction of $\mathbf{E}$ is opposite to the direction of increasing V. The surface charge densities at the conducting plates are obtained by using Eq. (3–72),

$$E_n = \mathbf{a}_n \cdot \mathbf{E} = \frac{\rho_s}{\epsilon}.$$

At the lower plate,

$$\mathbf{a}_n = \mathbf{a}_y, \qquad E_{n\ell} = -\frac{V_0}{d}, \qquad \rho_{s\ell} = -\frac{\epsilon V_0}{d}.$$

At the upper plate,

$$\mathbf{a}_n = -\mathbf{a}_y, \qquad E_{nu} = \frac{V_0}{d}, \qquad \rho_{su} = \frac{\epsilon V_0}{d}.$$

Electric field lines in an electrostatic field begin from positive charges and end in negative charges.

Solving boundary-value problems :

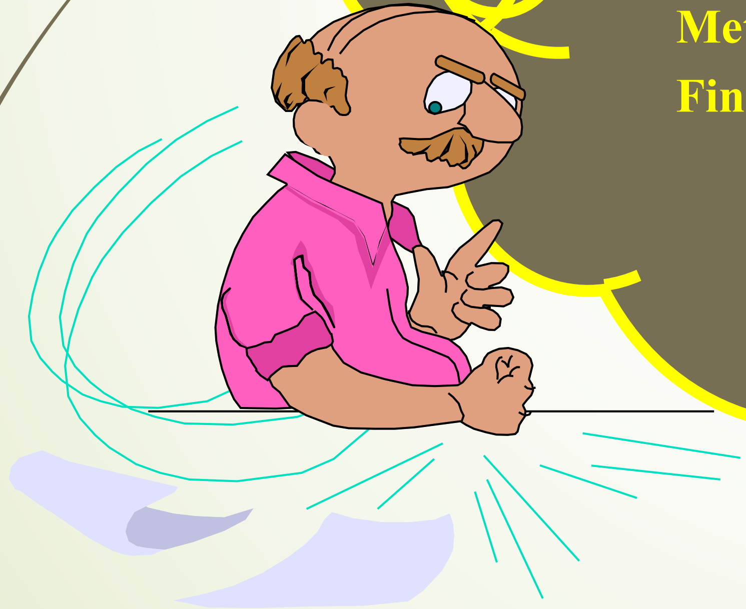
- Description of boundary value problems
- Solution of boundary value problem:

Method of images

Method of Separation of variables

Finite difference method

……..



4.3 Uniqueness of Electrostatic Solutions

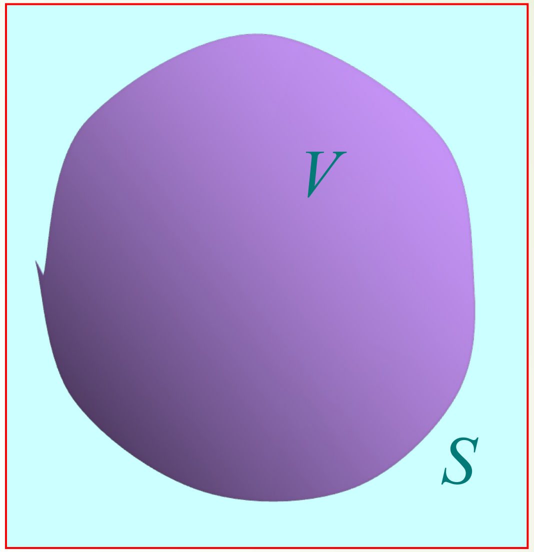
- There may be more than one method for one problem.
- Will these methods obtain different solutions?
- The uniqueness theorem

  A solution of Poisson's equation (of which Laplace's equation is a special case) that satisfies the given boundary conditions is a unique solution.

- The implication of the uniqueness theorem is *that a solution of an electrostatic problem satisfying its boundary conditions is the only possible solution, irrespective of the method by which the solution is obtained.*

The uniqueness theorem

Keep the equations of unknowns unchanged in the solution domain <u>*V*</u>; at the same time, keep the boundary value of φ or $\partial\varphi / \partial n$ unchanged on the boundary surface <u>*S*</u> ; then the solution of the Poisson's equations or the Laplace's equations in the solution domain <u>*V*</u> is unique



V: solution domain

S: Surface is surrounded by V

The importance of uniqueness theorem

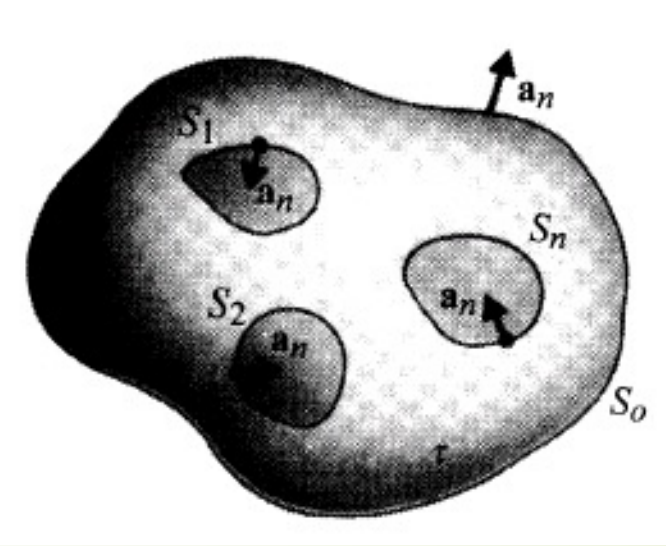
- **Give the condition of the boundary-value problems with a unique solution.**
- **Provide a theoretical basis for solving the field problems with different methods**
- **Provide a criterion for the correctness of the solution.**

Prove

- Suppose a volume τ is bounded outside by a surface S_0.
- Inside the closed surface S_0 there are a number of charged conducting bodies with surfaces S_1, S_2, S_3,…., S_n at specified potentials.
- Now assume that there are two solutions, V_1 and V_2 to Poisson's equation in τ,

$$\nabla^2 V_1 = -\frac{\rho}{\epsilon},$$

$$\nabla^2 V_2 = -\frac{\rho}{\epsilon}.$$



- **Also assume that both V_1 and V_2 satisfy the same boundary conditions on S_1, S_2, S_3,…., S_n and S_0. Define a new differential potential:**

$$V_d = V_1 - V_2.$$

- **We see that V_d satisfies Laplace's equation in τ.**

$$\nabla^2 V_d = 0.$$

- **Recalling the vector identity,**

$$\nabla \cdot (f\mathbf{A}) = f\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla f;$$

- **and letting $f = V_d$ and A = grad V_d; we have**

$$\nabla \cdot (V_d \nabla V_d) = V_d \nabla^2 V_d + |\nabla V_d|^2,$$

- where the first term on the right side vanishes. Integration of the Eq. over the volume τ yields

$$\oint_S (V_d \nabla V_d) \cdot \mathbf{a}_n \, ds = \int_\tau |\nabla V_d|^2 \, dv,$$

- where surface S consists of S_0 as well as S_1, S_2, S_3,…., S_n. Over the conducting boundaries, as well as over the large surface S_0, $V_\mathrm{d} = 0$. We have

$$\int_\tau |\nabla V_d|^2 \, dv = 0.$$

- Since the integrand is nonnegative everywhere,

$$\left|\nabla V_d\right| = 0.$$

- $V_\mathrm{d} = 0$ throughout the volume.

4.4 method of Images

4.4.1 The basic principle of image method

4.4.2 Point charge and conducting planes

4.4.3 Point charge and conducting sphere

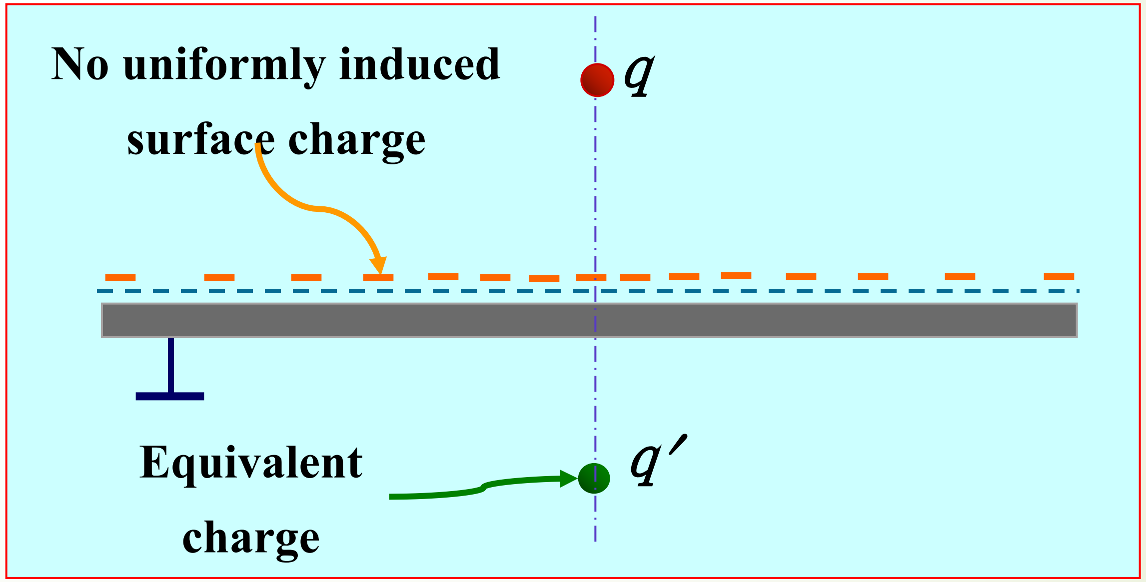
4.4.4 Line charge and conducting cylinder

4.4.1 The basic principle of image method

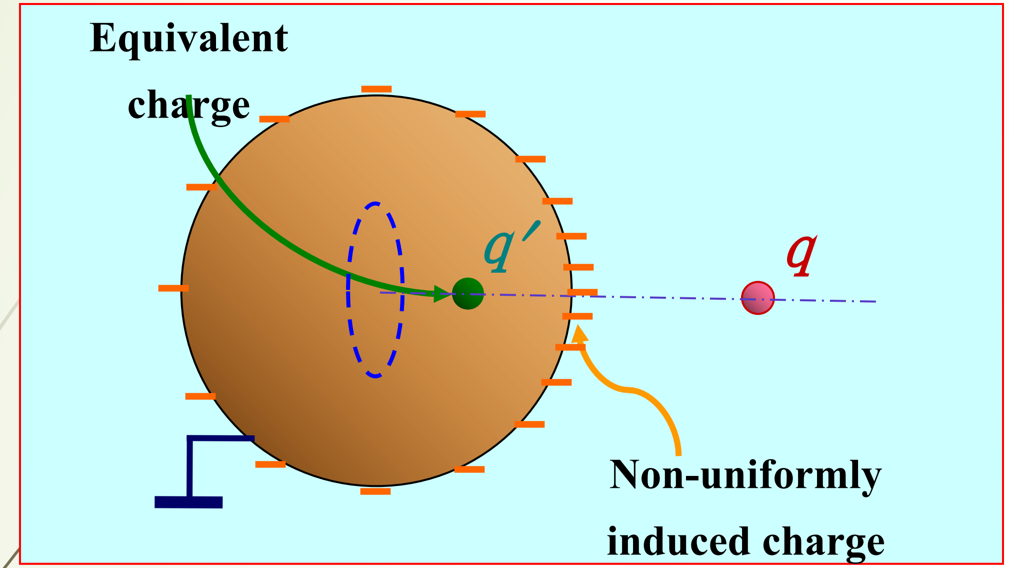
1. Introduce the questions

- **Several examples**

A point charge near the grounded conducting plane.



- A point charge near the grounded conducting sphere.



- A line charge near the grounded conducting cylinder, which is similar with the one above, but the equivalent charge is line charge。

- **Question: :** Does this equivalent charge exist?
  Is this equivalent method reasonable?

2. Principle of the method of image

- **Essence:** The effect of the boundary is replaced by one or several equivalent charges, and the original inhomogeneous region with a boundary becomes an infinite homogeneous space.

- **Key:** To determine the number, the positions and the values of the image charges.

- **Restriction:** These image charges may be determined only for some special boundaries and charges with certain distributions.

2. Principle of the method of image

- **Purpose:** Change the complex boundary-value problem into the problem with infinite single medium space.

- **Basis:** The principle of **uniqueness**. Therefore, these charges should not change the original boundary conditions. These equivalent charges are at the image positions of the original charges, and are called **image charges**, and this method is called the method of images.

3. The key points of the application of image method

- Determination of the image charge

  The number, position and value of the image charge

  ——"Three key elements"

- Explicit “the effective field” of equivalent solution

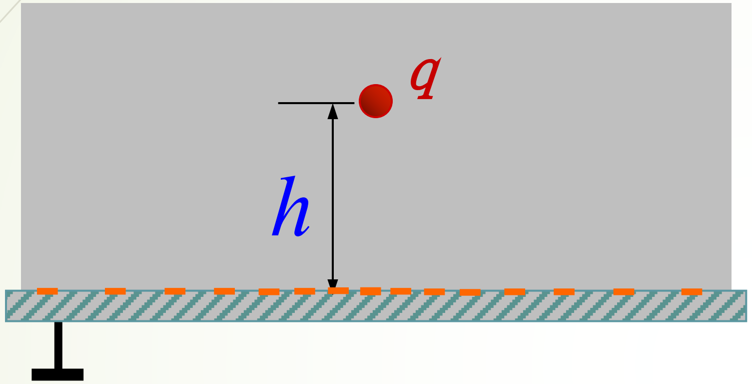
4. Two principles of determining image charge

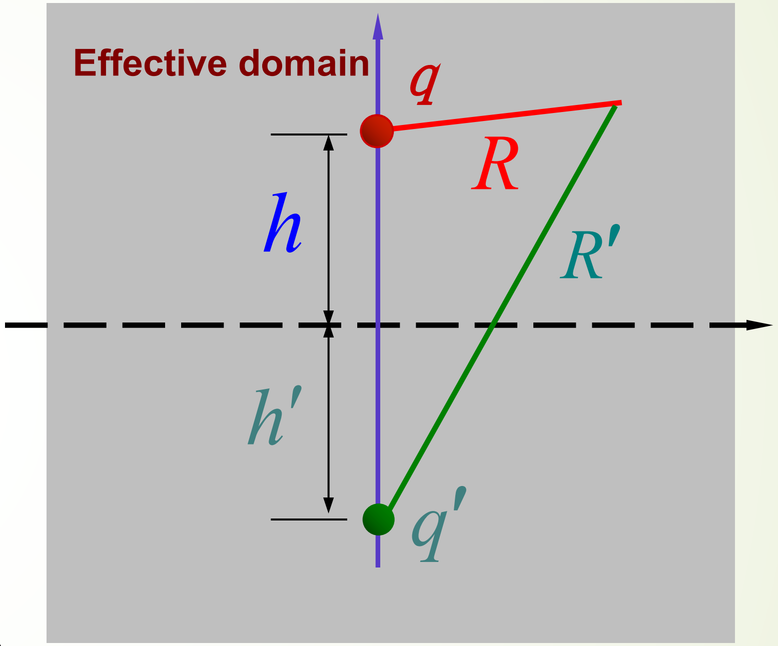
- The image charge must be located outside the solution domain

- The choice of the number, the position and the charge quantity of the image charge is to keep the boundary conditions of the problem unchanged.

4.4.2 Point charge and conducting planes

1. The image of the point charge on an infinite grounded plane conductor

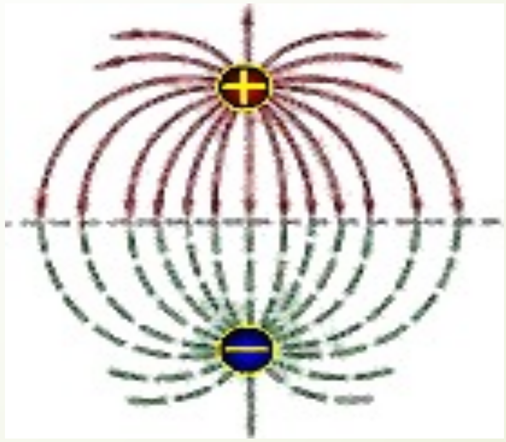




The image charge $q' = -q, \quad h' = h,$

The potential functions $\varphi = \dfrac{q}{4\pi\varepsilon}\left(\dfrac{1}{R} - \dfrac{1}{R'}\right) \qquad (z \geq 0)$

When $z = 0$, $R = R' \Rightarrow \varphi\big|_{z=0} = 0$



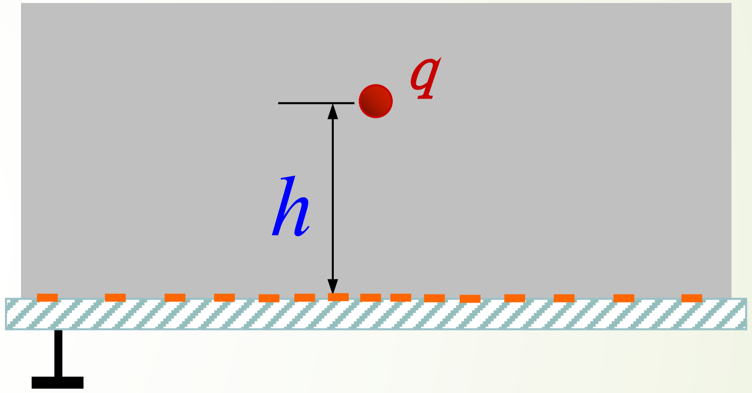
It satisfies the original boundary value problem, and the result is correct!

• **Solving the induced surface charge density and the total induced charge of the grounded conducting plane**

$$\varphi(x,y,z)=\frac{q}{4\pi\varepsilon}\left[\frac{1}{\sqrt{x^2+y^2+(z-h)^2}}-\frac{1}{\sqrt{x^2+y^2+(z+h)^2}}\right]\quad(z\geq 0)$$

The induced surface charge density in the conductor plane is

$$\rho_S=-\varepsilon\left.\frac{\partial\varphi}{\partial z}\right|_{z=0}=-\frac{qh}{2\pi(x^2+y^2+h^2)^{3/2}}$$



The total induced charge on the conducting plane is

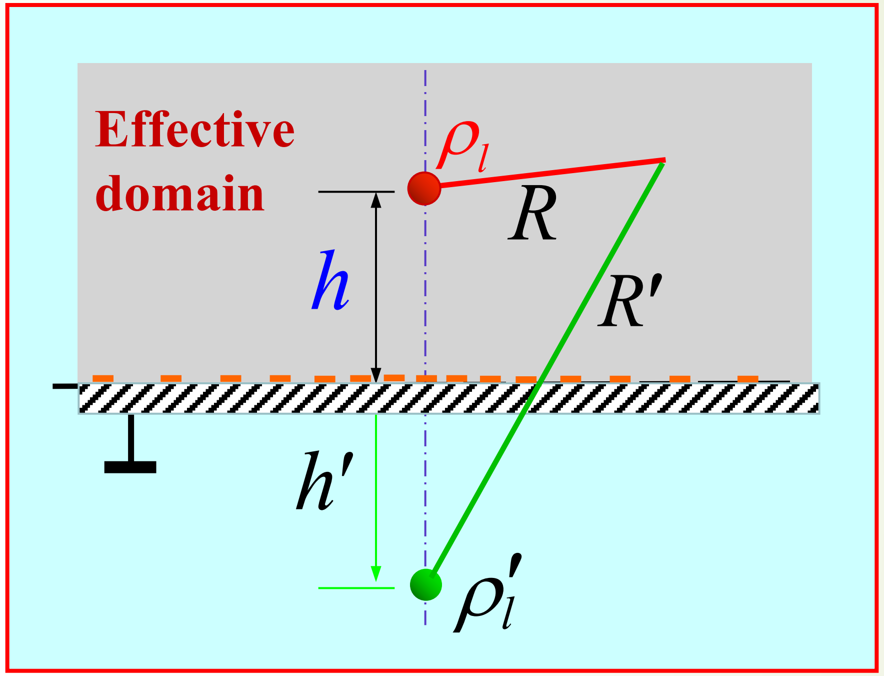
$$q_{in}=\iint_S \rho_S\,\mathrm{d}S=-\frac{qh}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{\mathrm{d}x\,\mathrm{d}y}{(x^2+y^2+h^2)^{3/2}}$$

$$=-\frac{qh}{2\pi}\int_0^{2\pi}\int_0^{\infty}\frac{\rho\,\mathrm{d}\rho\,\mathrm{d}\phi}{(\rho^2+h^2)^{3/2}}=-q$$

2. The image of the line charge on an infinite grounded plane conductor

Boundary Value

$$\begin{cases} \nabla^2 \varphi = -\dfrac{\rho_l}{\varepsilon}\delta(x, z-h) & (z>0) \\ \varphi = 0, & z = 0 \end{cases}$$



The images of line charge :

$$\rho_l' = -\rho_l, \quad h' = h$$

The potential functions :

$$\varphi = \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{R'}{R}\right) \quad (z \geq 0)$$

When $z = 0$, $r = r'$ $\Rightarrow$ $\varphi = 0$

It satisfies the original boundary value problems ,The results are correct!

3. The image of point charge between the Intersecting semi-infinite grounded conducting plane

q for plane 1:

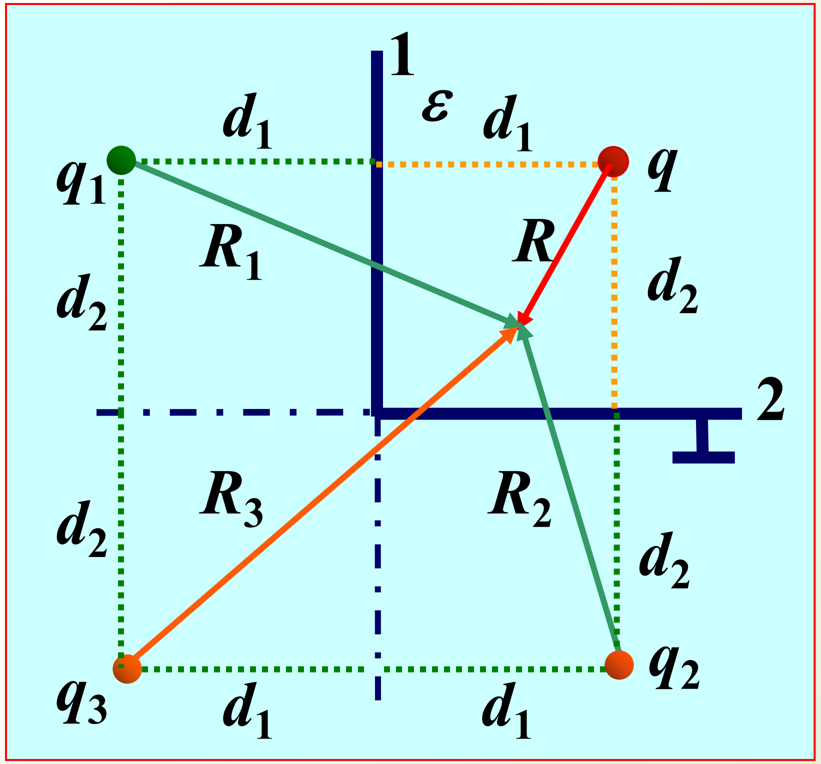
The image charge is $q_1 = -q$, located at $(-d_1, d_2)$

q for plane 2:

The image charge is $q_2 = -q$, located at $(d_1, -d_2)$

q_1 for plane 2 and q_2 for plane 1:

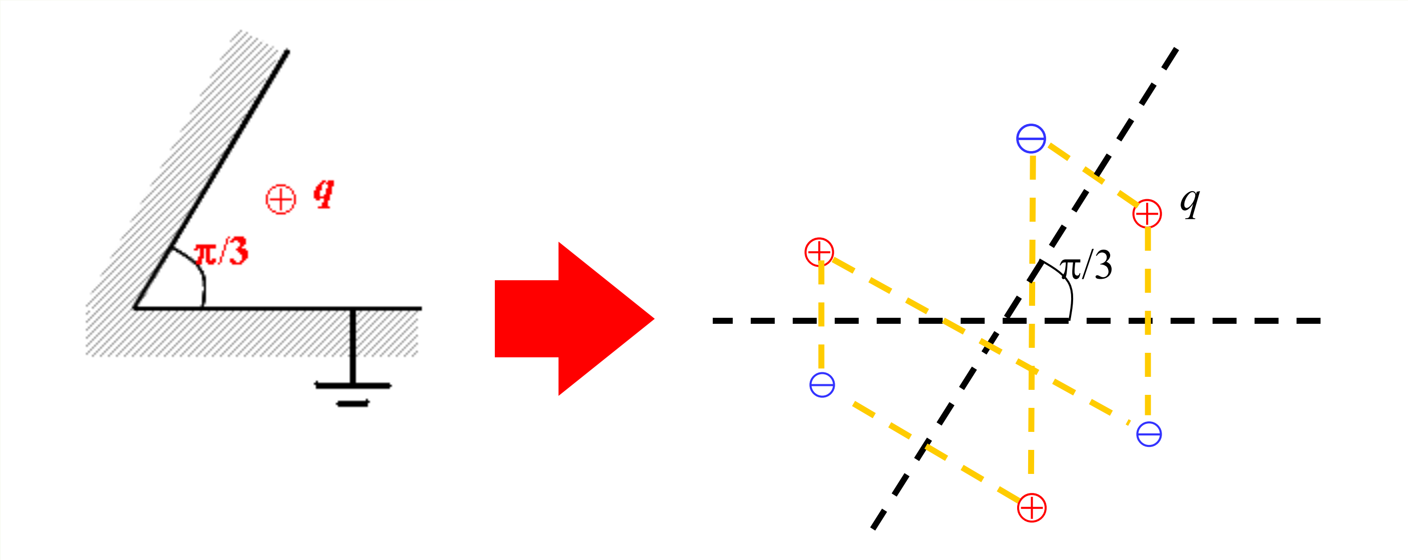
The image charge is $q_3 = q$, located at $(-d_1, -d_2)$



The solution of the potential function is :

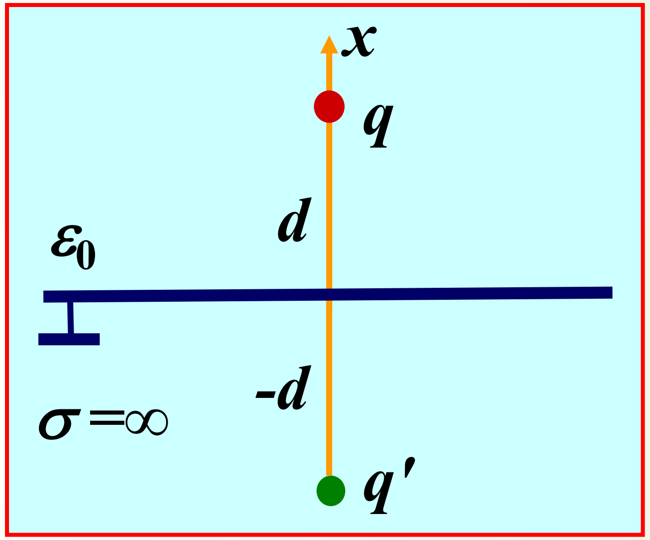
$$\varphi = \frac{q}{4\pi\varepsilon}\left(\frac{1}{R} - \frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3}\right)$$

• For the semi-infinite wedge conducting boundary, the method of images is also applicable. However, the images can be found only for conducting wedges with angle given by $\frac{\pi}{n}$ where n is an integer. In order to keep the wedge boundary at zero-potential, 2n-1 image charges are required.



When an infinite line charge is nearby an infinite conducting plane, the method of images can be applied as well, based on the principle of superposition.

Ex: The distance between a point charge q and an infinite conducting plane is d , If moves it to infinity, how much work needs to be done?



Solution:

$$\vec{E}'(x) = E_x(x)\vec{i} = \frac{-q}{4\pi\varepsilon_0(2x)^2}\vec{i}$$

$$W_e = q\int_d^\infty \vec{E}'(x)\cdot \mathrm{d}\vec{x}$$

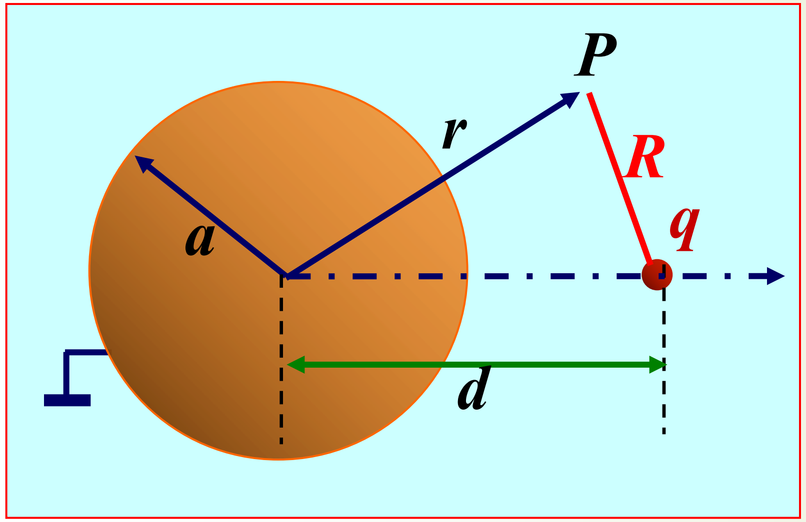
$$= -\frac{q^2}{4\pi\varepsilon_0(2)^2}\int_d^\infty \frac{1}{x^2}\mathrm{d}x = -\frac{q^2}{16\pi\varepsilon_0 d}$$

$$W = -W_e = \frac{q^2}{16\pi\varepsilon_0 d}$$

4.4.3 Point charge and conducting sphere

1. The image of the point charge in the grounded conducting sphere

$$\varphi = \frac{1}{4\pi\varepsilon}\left(\frac{q}{R}+\frac{q'}{R'}\right)$$

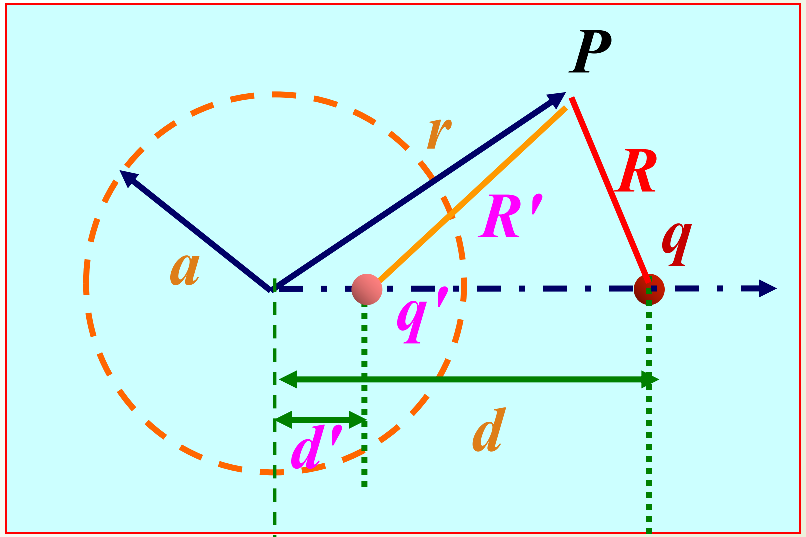


Question: $d' = ?$ $q' = ?$

Method: ?

Boundary conditions!

$$\varphi(a) = 0$$



$$\left.\left(\frac{q}{R}+\frac{q'}{R'}\right)\right|_{r=a}=0$$

$$\Rightarrow \left.\frac{R'}{R}\right|_{r=a}=-\frac{q'}{q}=\text{Constant}$$

$$\Rightarrow \frac{a}{d}=\frac{d'}{a}$$

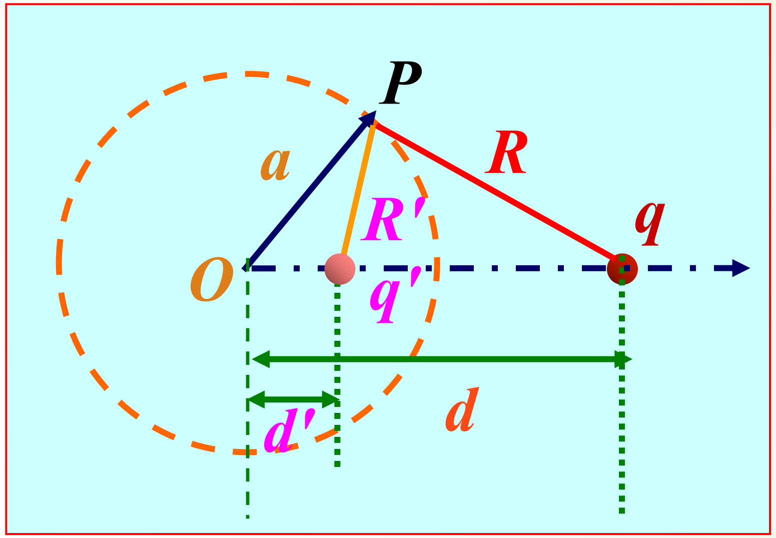
$$\Rightarrow d'=\frac{a^2}{d}$$

Position of the image charge

$$-\frac{q'}{q}=\left.\frac{R'}{R}\right|_{r=a}=\frac{a}{d}$$

$$\Rightarrow q'=-\frac{a}{d}q$$

Charge of the image charge



- **Find the induced surface charge density and the total induced charge on the conducting sphere**

$$\varphi = \frac{q}{4\pi\varepsilon}\left[\frac{1}{\sqrt{r^2+d^2-2rd\cos\theta}} - \frac{a}{\sqrt{d^2r^2+a^4-2rda^2\cos\theta}}\right] \quad (r \geq a)$$

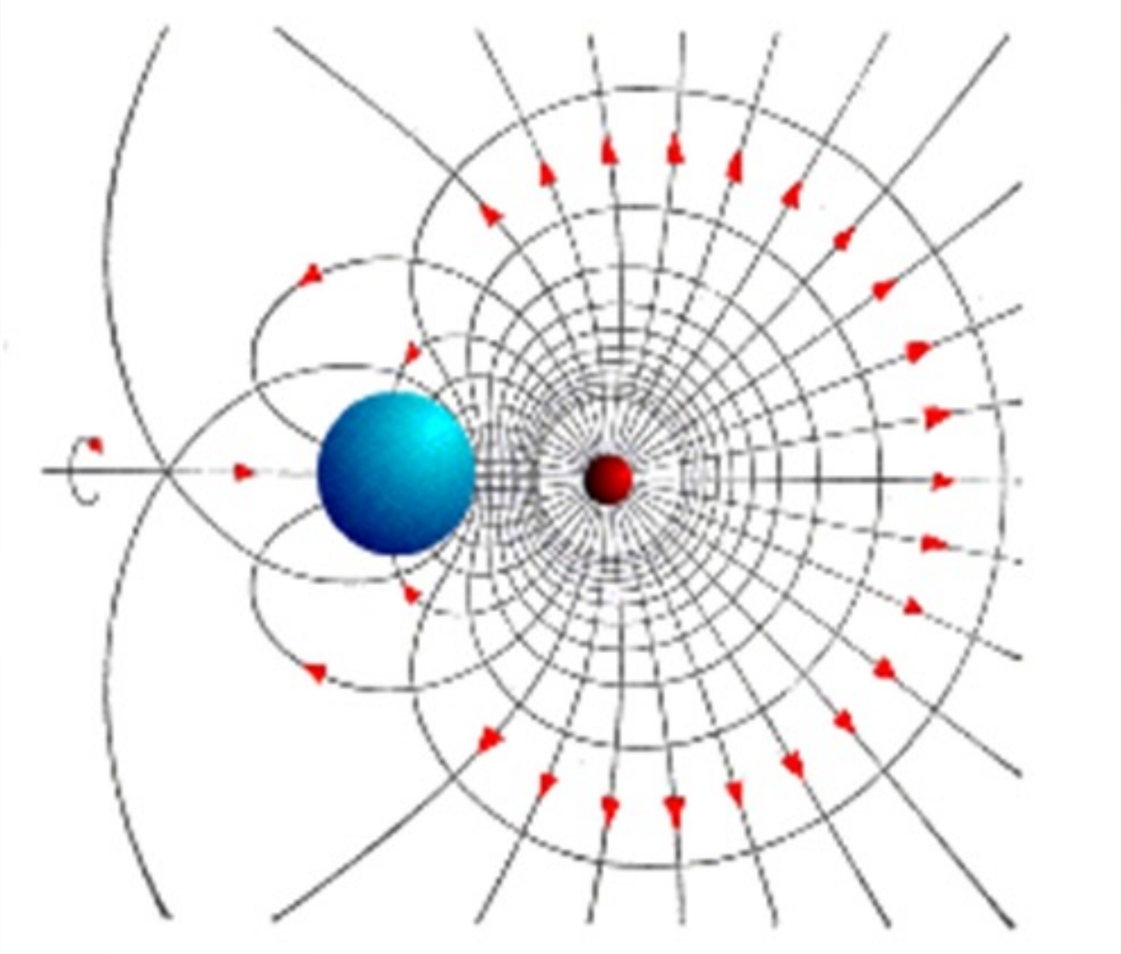
The induction surface charge density on the sphere is

$$\rho_S = -\varepsilon\left.\frac{\partial\varphi}{\partial r}\right|_{r=a} = -\frac{q(d^2-a^2)}{4\pi a(a^2+d^2-2ad\cos\theta)^{3/2}}$$

The total induced charge on the conducting sphere is

$$q_{in} = \oint_S \rho_S \, dS = -\frac{q(d^2-a^2)}{4\pi a}\int_0^{2\pi}\int_0^{\pi}\frac{a^2\sin\theta \, d\theta \, d\phi}{(a^2+d^2-2ad\cos\theta)^{3/2}} = -\frac{a}{d}q$$

- **The total induced charge on the surface of the conducting sphere is equal to the image charge of placement.**

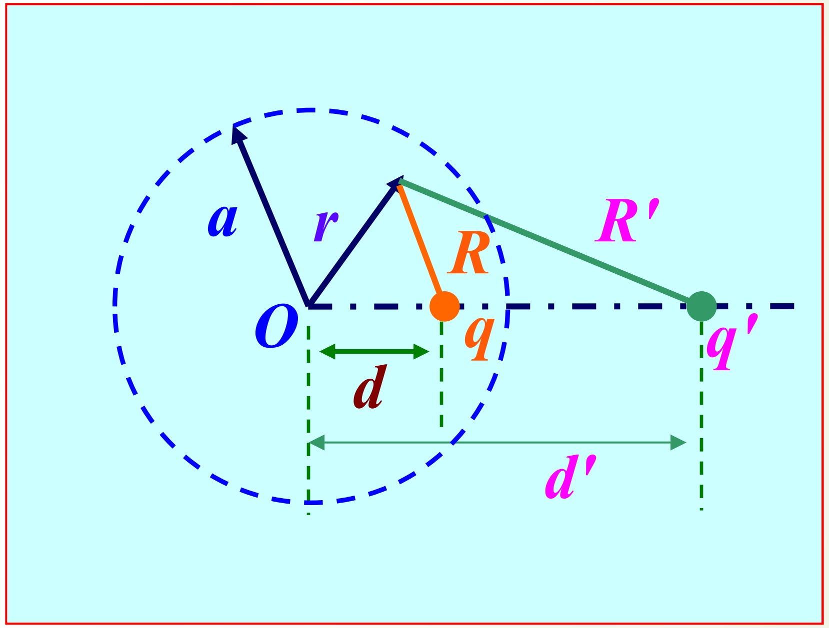


Electric Field for a point charge located near a grounded conductor sphere.

The image charge of a point charge located inside the grounded hollow conducting spherical shell

$$q' = -\frac{a}{d}q, \quad d' = \frac{a^2}{d}$$

$$\varphi = \frac{1}{4\pi\varepsilon}\left(\frac{q}{R} + \frac{q'}{R'}\right) \quad (r \le a)$$



- $|q'| > |q|$
- It has nothing to do with the outer radius b（Why? ）

- **Find the induced charge density and the total induced charge on the inner surface of the conductor spherical shell**

$$\varphi = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{r^2+d^2-2rd\cos\theta}} - \frac{a}{\sqrt{r^2d^2+a^4-2ra^2d\cos\theta}}\right] \quad (r \le a)$$

The induced surface charge density is :

$$\rho_S = -\varepsilon_0 \left.\frac{\partial \varphi}{\partial r}\right|_{r=a} = -\frac{q(a^2-d^2)}{4\pi a(a^2+d^2-2ad\cos\theta)^{3/2}}$$

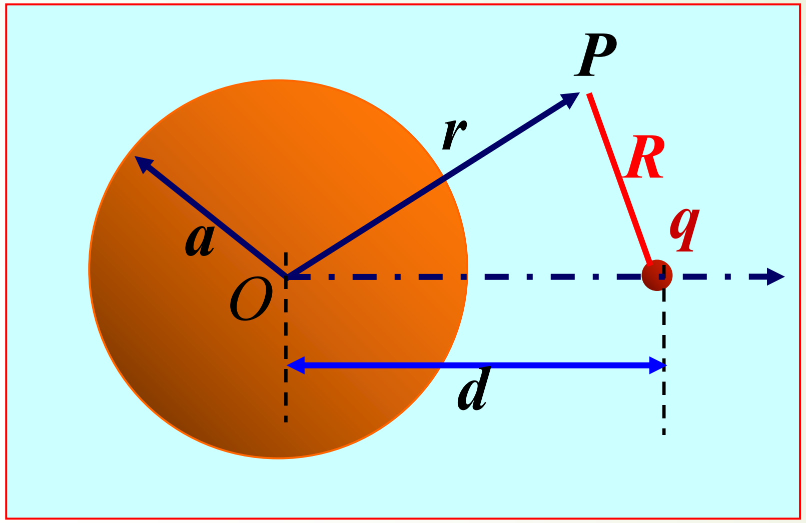
The total induced charge on the inner surface of the conducting spherical shell is

$$q_{in} = \oint_S \rho_S \,\mathrm{d}S = -\frac{q(a^2-d^2)}{4\pi}\int_0^{2\pi}\int_0^{\pi} \frac{a\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi}{(a^2+d^2-2ad\cos\theta)^{3/2}} = -q$$

Is equivalent charge always equal to the total of induced charge？

NO！

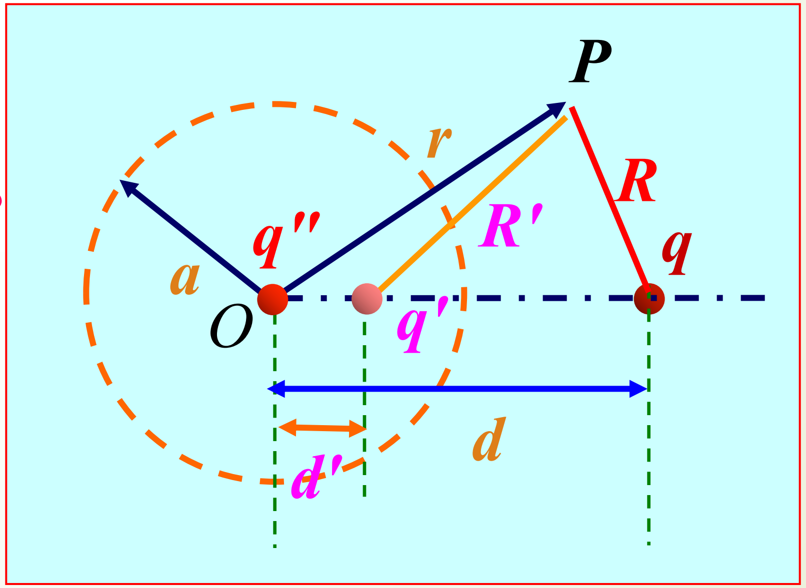
2 . The image of point charge on the ungrounded conducting sphere



- The ungrounded conducting sphere :
  - An non-zero equal potential body.
  - The total induced charge on the spherical surface is zero.
  - The surface distribution of the induced charge is :

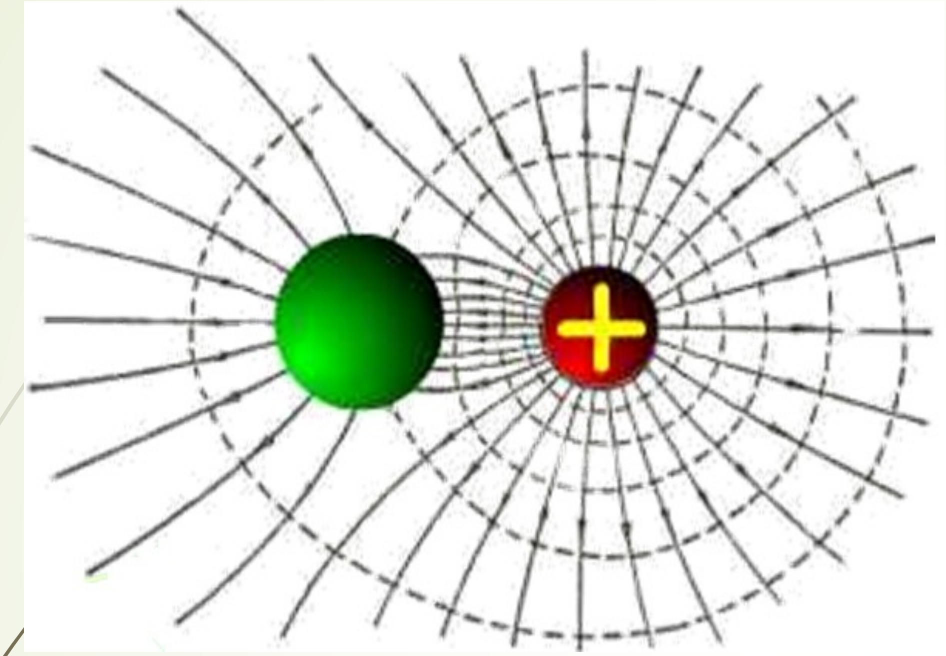
Is negative charge distribution the same as the grounded sphere?

Is positive charges uniformly distributed?

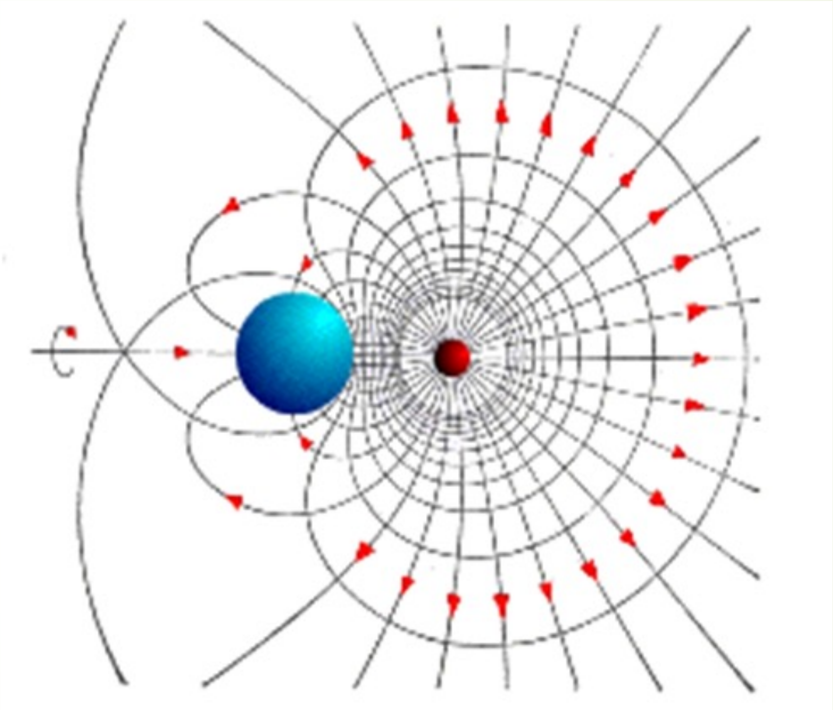


$$q' = -\frac{a}{d}q, \quad d' = \frac{a^2}{d}, \quad q'' = -q'$$

$$\varphi = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{R} + \frac{q'}{R'} + \frac{q''}{r}\right)$$

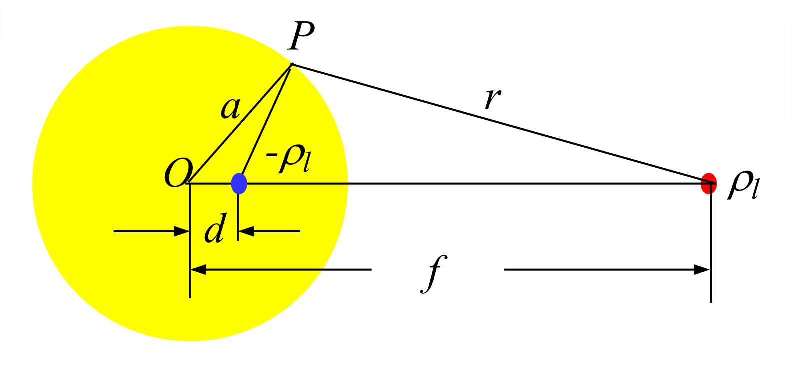


Electric Field for a point charge located near an ungrounded conductor sphere.



Electric Field for a point charge located near a grounded conductor sphere.

4.4.4 Line charge and conducting cylinder



An **image** line charge $-\rho_l$ is put in to represent the charge on the cylinder and placed **parallel** to it, at a distance d from the axis.

The **electric field intensity** produced by an infinite line charge of density ρ_l given by

$$\boldsymbol{E} = \frac{\rho_l}{2\pi\varepsilon r}\boldsymbol{e}_r$$

The electric potential with respect to **a reference point** r_0 at a distance r from the line electric charge is

$$\varphi = -\int_{r_0}^{r} E\mathrm{d}r = \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{r_0}{r}\right)$$

If r_0 is also taken as the reference point for the electric potential produced by the image line charge $-\rho_l$, then ρ_l and $-\rho_l$ produce an electric potential at a point *P* on the surface of the cylinder given by

$$\varphi_P = \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{r_0}{r}\right) - \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{r_0}{r'}\right) = \frac{\rho_l}{2\pi\varepsilon}\ln\left(\frac{r'}{r}\right)$$

Since the conducting cylinder is an equipotential body, in order to satisfy this boundary condition the ratio $\frac{r'}{r}$ must be a constant.

Let $\frac{r'}{r} = \frac{a}{f} = \frac{d}{a}$, we find

$$d = \frac{a^2}{f}$$

Using image method to solve the problems

1. Grounded conducting plane
2. Grounded conducting sphere (including the inside and outside of the sphere)
3. Ungrounded conducting spherical (including the inside and outside of the sphere)

Question:

- Find the potential function of area with a point charge on the ungrounded infinite conducting plane ?

Analysis method summary

Methods have been learned and we can solve the problems

1. **The problem of infinite single medium space （ 1D,2D,3D problems)**

- ✓ **Field—source directly integration method**
- ✓ **Integral equation method （Integral form of the Maxwell’s equations ）**
- ● **Differential equation method (differential form of Maxwell’s equations, Poisson’s equations)**

2. **Single / non-single medium space problem (1D problems)**

- ✓ **Gauss's law and Ampere's Law (the integral equation is simplified to the algebraic equation)**
- ✓ **Poisson’s equations (partial differential equation is simplified to ordinary differential equation).**

3. **Higher dimensional problems in non-single medium space**

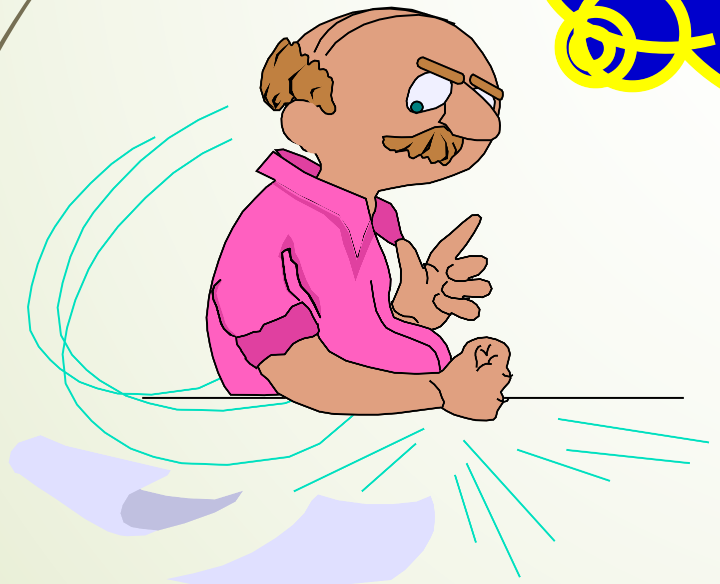
   √ The method of images

❖ Boundary-value problem:

In the given boundary conditions, the solution of the Poisson's equations or Laplasce's equations

Solving boundary value problems :

- Description of boundary value problems
- Solution of boundary value problems



4.5 Boundary-Value Problems in Cartesian Coordinates

- **Method of image is useful in solving problems involving free charges near conducting boundaries that are geometrically simple.**

- **Problems consists of a system of conductors maintained at specified potentials and with no isolated free charges, requires the solution of Laplace's equation.**

- **Problems governed by partial differential equations with prescribed boundary conditions are called *boundary-value problems.***

Types of boundary value

- **The first boundary-value problems (or Dirichlet problems)**

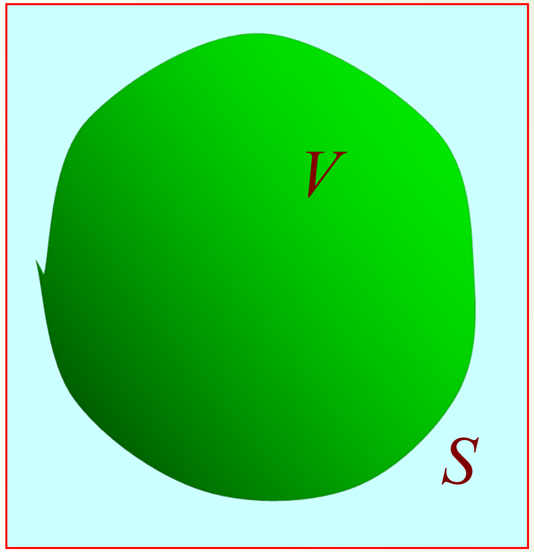
  Given $\varphi|_S = f_1(S)$

- **The second-boundary value problems (or Newman problems)**

  Given $\left.\dfrac{\partial \varphi}{\partial n}\right|_S = f_2(S)$

- **The third boundary-value problems (or mixed boundary-value problems)**

  Given $\varphi|_{S_1} = f_{11}(S_1)$ 、 $\left.\dfrac{\partial \varphi}{\partial n}\right|_{S_2} + f_{22}(S_2) = f_{12}(S_2)$



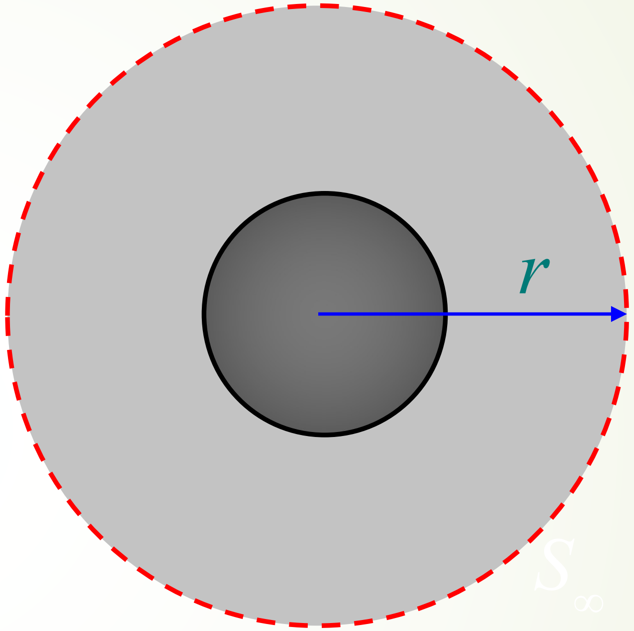
V: solution domain

S: Surface is surrounded by V

Natural boundary condition

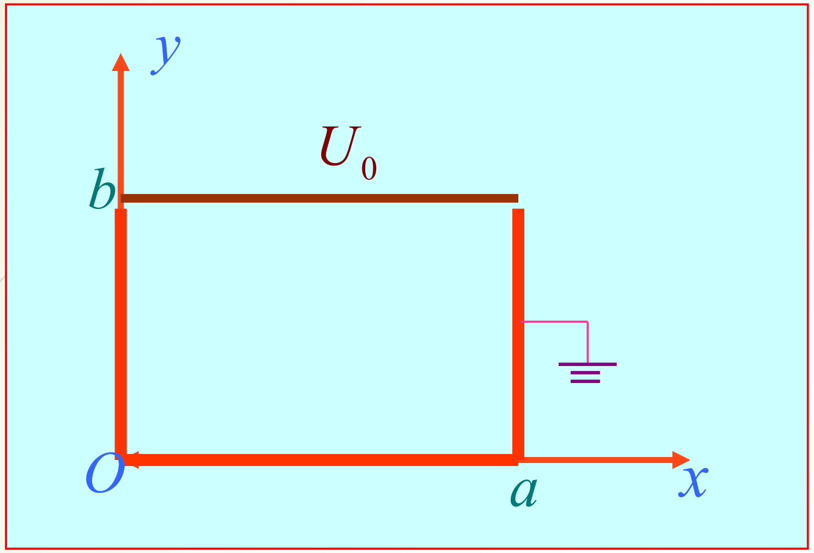
(open space)

$$\lim_{r\to\infty} r\varphi = \text{finite value}$$



Requirements: master the mathematical description method to solve complex problems with the idea of boundary-value problems

Ex:



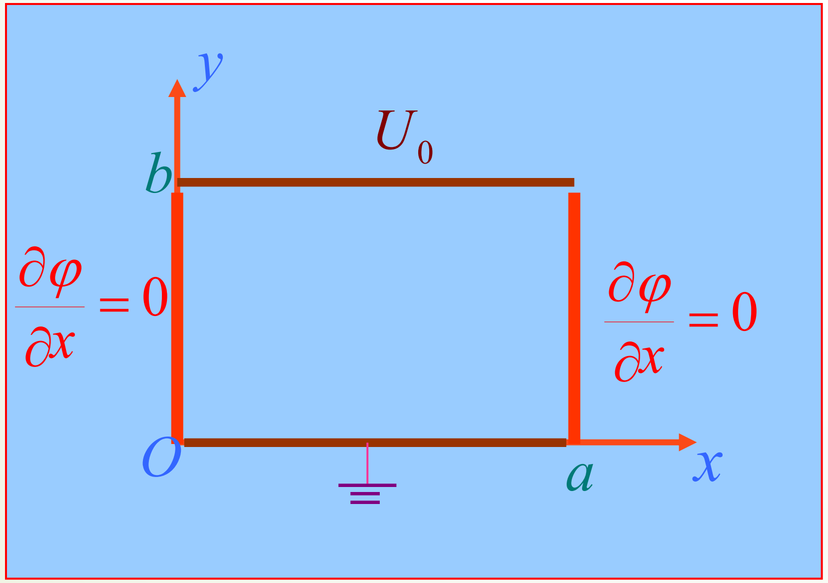
$$\frac{\partial^2 \varphi}{\partial x^2}+\frac{\partial^2 \varphi}{\partial y^2}=0$$

$$\varphi(0,y)=\varphi(a,y)=0$$

$$\varphi(x,0)=0,\quad \varphi(x,b)=U_0$$

（ The first boundary value problems ）

Ex:



$$\frac{\partial^2 \varphi}{\partial x^2}+\frac{\partial^2 \varphi}{\partial y^2}=0$$

$$\left.\frac{\partial \varphi}{\partial x}\right|_{x=0}=0,\quad \left.\frac{\partial \varphi}{\partial x}\right|_{x=a}=0$$

$$\varphi(x,0)=0,\quad \varphi(x,b)=U_0$$

（ The third boundary value problems ）

- Different specified boundary conditions will require the choice of **different potential functions**, but the procedure of solving these types of problems is the **same**-the **method of separation of variables**;

- The solutions of Laplace's equation are called **harmonic functions**.

The basic principle of method of separation of variables

- **Method of Separation of variables is a general method to solve boundary-value problems**

- **The basic idea of method of separation of variables :**

  The dependency relation of the unknown function on several variables (high dimension problem) is changed into Separate dependent relations of each variable (variables can be separated), so that high dimensional problem can be transformed into one dimensional problem.

- **The theoretical basis of the method of separation of variables is uniqueness theorem**

In rectangular coordinate system, Laplace's Equation for electric potential is

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

Let $\varphi(x,\ y,\ z) = X(x)Y(y)Z(z)$

Substituting it into the above equation, and dividing both sides by $X(x)Y(y)Z(z)$, we have

$$\frac{1}{X}\frac{\mathrm{d}^2 X}{\mathrm{d}x^2} + \frac{1}{Y}\frac{\mathrm{d}^2 Y}{\mathrm{d}y^2} + \frac{1}{Z}\frac{\mathrm{d}^2 Z}{\mathrm{d}z^2} = 0$$

Where each term involves only one variable. The only way the equation can be satisfied is to have each term equal to a constant.

Let these constants be $-k_x^2,\ -k_y^2,\ -k_z^2$, and we have

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0 \qquad \frac{d^2 Y}{dy^2} + k_y^2 Y = 0 \qquad \frac{d^2 Z}{dz^2} + k_z^2 Z = 0$$

where k_x , k_y , k_z are called the separation constants, and they could be real or imaginary numbers.

The three separation constants are not independent of each other, and they satisfy the following equation

$$k_x^2 + k_y^2 + k_z^2 = 0$$

The three-dimensional partial differential equation is separated to three ordinary differential equations, and the solutions of the ordinary differential equations are easier to obtain.

The solution of the equation for the variable x can be written as

$$X(x) = A e^{-jk_x x} + B e^{jk_x x} \quad \text{or} \quad X(x) = C \sin k_x x + D \cos k_x x$$

where A, B, C, D are the constants to be determined.

The separation constants could be imaginary numbers. If k_x is an imaginary number, written as $k_x = jk$, then the equation becomes

$$X(x) = Ae^{kx} + Be^{-kx} \quad \text{or} \quad X(x) = C\sinh kx + D\cosh kx$$

The solutions of the equations for the variables *y* and *z* have the same forms. The product of these solutions gives the solution of the original partial differential equation.

It is very important to select the forms of the solutions, which depend on the given boundary conditions.

The constants in the solutions are also related to the boundary conditions.

TABLE 4–1
Possible Solutions of $X''(x) + k_x^2 X(x) = 0$

k_x^2	k_x	$X(x)$	Exponential forms† of $X(x)$
0	0	$A_0 x + B_0$	
+	k	$A_1 \sin kx + B_1 \cos kx$	$C_1 e^{jkx} + D_1 e^{-jkx}$
−	jk	$A_2 \sinh kx + B_2 \cosh kx$	$C_2 e^{kx} + D_2 e^{-kx}$

† The exponential forms of $X(x)$ are related to the trigonometric and hyperbolic forms listed in the third column by the following formulas:

$$e^{\pm jkx} = \cos kx \pm j \sin kx, \quad \cos kx = \tfrac{1}{2}(e^{jkx} + e^{-jkx}), \quad \sin kx = \frac{1}{2j}(e^{jkx} - e^{-jkx});$$

$$e^{\pm kx} = \cosh kx \pm \sinh kx, \quad \cosh kx = \tfrac{1}{2}(e^{kx} + e^{-kx}), \quad \sinh kx = \tfrac{1}{2}(e^{kx} - e^{-kx}).$$

In the Cartesian coordinate system， If the potential function has nothing to do with Z， then the Laplace’s equations is

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0$$

The $\varphi(x, y)$ is expressed as the product of two one-dimensional functions $X(x)$ and $Y(y)$， that is

$$\varphi(x, y) = X(x)Y(y)$$

It will be substituted into the Laplace’s equation， then

$$Y(y)\frac{\mathrm{d}^2 X(x)}{\mathrm{d}x^2} + X(x)\frac{\mathrm{d}^2 Y(y)}{\mathrm{d}y^2} = 0$$

Then divided by $X(x)\,Y(y)$， next

$$\frac{1}{X(x)}\frac{\mathrm{d}^2 X(x)}{\mathrm{d}x^2} = -\frac{1}{Y(y)}\frac{\mathrm{d}^2 Y(y)}{\mathrm{d}y^2} = \lambda$$

Separation constant

If $\lambda = -k^2$， then

$$\frac{\mathrm{d}^2 X(x)}{\mathrm{d}x^2} + k^2 X(x) = 0 \qquad \frac{\mathrm{d}^2 Y(y)}{\mathrm{d}y^2} - k^2 Y(y) = 0$$

When $k = 0 \Rightarrow X_0(x) = A_0 x + B_0 \qquad Y_0(y) = C_0 y + D_0$

$$\Rightarrow \varphi_0(x,y) = X_0(x)Y_0(y) = (A_0 x + B_0)(C_0 y + D_0)$$

when $k = k_n \neq 0 \Rightarrow X_n(x) = A_n \sin(k_n x) + B_n \cos(k_n x)$

$$Y_n(y) = C_n \sinh(k_n y) + D_n \cosh(k_n y)$$

$$\Rightarrow \varphi_n(x,y) = X_n(x) Y_n(x)$$

$$= [A_n \sin(k_n x) + B_n \cos(k_n x)][C_n \sinh(k_n y) + D_n \cosh(k_n y)]$$

If all possible $\varphi(x, y)$ are linearly superimposed., we can get a general solution of the potential functions，that is

$$\varphi(x,y) = (A_0 x + B_0)(C_0 y + D_0)$$

$$+ \sum_{n=1}^{\infty} [A_n \sin(k_n x) + B_n \cos(k_n x)][C_n \sinh(k_n y) + D_n \cosh(k_n y)]$$

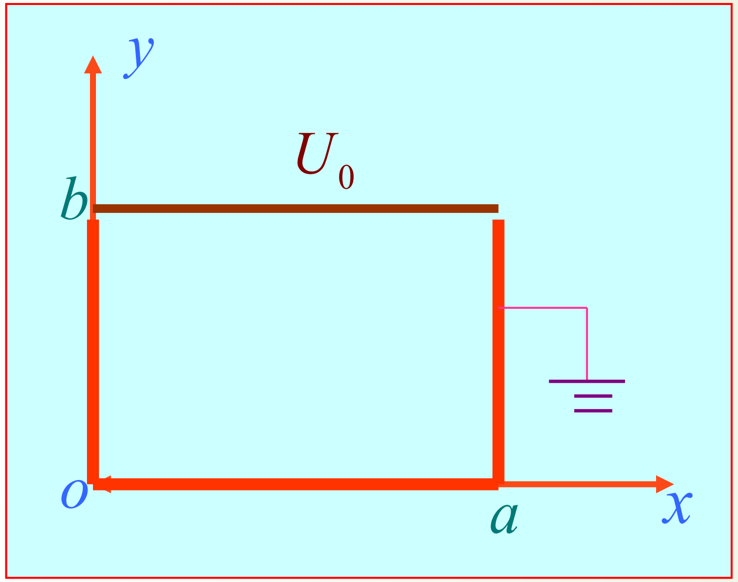
- **Separation constant and undetermined coefficients in the general solution is determined by the given boundary conditions.**

If $\lambda = k^2$,Similarly , we can get that

$$\varphi(x,y) = (A_0 x + B_0)(C_0 y + D_0)$$

$$+ \sum_{n=1}^{\infty} [A_n \sinh(k_n x) + B_n \cosh(k_n x)][C_n \sin(k_n y) + D_n \cos(k_n y)]$$

Ex: **A cover plate is on the infinite rectangular mental conductor slot, which are isolated with each other. The potential of the cover plate is U0, mental conductor slot is grounded. Find the potential distribution in the conductor slot.**



Solution:

$$\frac{\partial^2 \varphi}{\partial x^2}+\frac{\partial^2 \varphi}{\partial y^2}=0$$

$$\varphi(0,y)=\varphi(a,y)=0 \quad (0\le y\le b)$$

$$\varphi(x,0)=0,\ \varphi(x,b)=U_0 \quad (0\le x\le a)$$

Since $\varphi(0,y)=0$、 $\varphi(a,y)=0$， **So the general solution is**

$$\varphi(x,y)=(A_0x+B_0)(C_0y+D_0)$$

$$+\sum_{n=1}^{\infty}[A_n\sin(k_nx)+B_n\cos(k_nx)][C_n\sinh(k_ny)+D_n\cosh(k_ny)]$$

Determinate the coefficient

$$\varphi(0,y)=0 \Rightarrow B_0(C_0 y + D_0) + \sum_{n=1}^{\infty} B_n[C_n\sinh(k_n y) + D_n\cosh(k_n y)] = 0$$

$$\Rightarrow B_0 = 0,\ B_n = 0$$

$$\Rightarrow \varphi(x,y) = A_0 x(C_0 y + D_0) + \sum_{n=1}^{\infty} A_n\sin(k_n x)[C_n\sinh(k_n y) + D_n\cosh(k_n y)]$$

$$\varphi(a,y)=0 \Rightarrow A_0 a(C_0 y + D_0) + \sum_{n=1}^{\infty} A_n\sin(k_n a)[C_n\sinh(k_n y) + D_n\cosh(k_n y)] = 0$$

$$\Rightarrow A_0 a = 0,\ A_n\sin(k_n a) = 0 \qquad \Rightarrow A_0 = 0,\ \sin(k_n a) = 0$$

$$\Rightarrow A_0 = 0,\ k_n = \frac{n\pi}{a}$$

$$\Rightarrow \varphi(x,y) = \sum_{n=1}^{\infty}\sin\left(\frac{n\pi x}{a}\right)\left[C_n\sinh\left(\frac{n\pi y}{a}\right) + D_n\cosh\left(\frac{n\pi y}{a}\right)\right]$$

$$\varphi(x,0) = 0 \;\Rightarrow\; \sum_{n=1}^{\infty} A_n D_n \sin\left(\frac{n\pi x}{a}\right) = 0 \;\Rightarrow\; D_n = 0$$

$$\Rightarrow\; \varphi(x,y) = \sum_{n=1}^{\infty} A_n C_n \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi y}{a}\right) = \sum_{n=1}^{\infty} A_n' \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi y}{a}\right), \quad A_n' = A_n C_n$$

$$\varphi(x,b) = U_0 \;\Rightarrow\; \sum_{n=1}^{\infty} A_n' \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi b}{a}\right) = U_0$$

Expand the U_0 using $\left\{\sin\left(\frac{n\pi x}{a}\right)\right\}$ as Fourier series on (0,a)

$$U_0 = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi x}{a}\right)$$

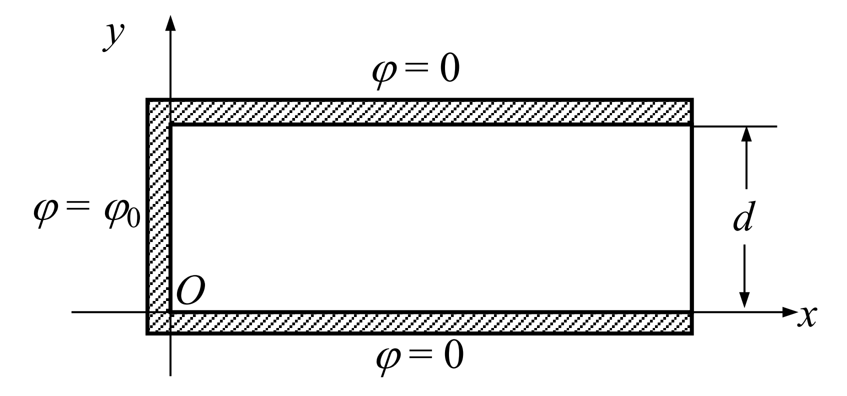
where $$f_n = \frac{2}{a}\int_0^a U_0 \sin\left(\frac{n\pi x}{a}\right) dx = \begin{cases} \dfrac{4U_0}{n\pi} & n = 1,3,5,\ldots \\ 0 & n = 2,4,6,\ldots \end{cases}$$

then $$\sum_{n=1}^{\infty} A'_n \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi b}{a}\right) = U_0$$

$$A'_n = \frac{f_n}{\sinh\left(\frac{n\pi b}{a}\right)} = \begin{cases} \dfrac{4U_0}{n\pi \sinh(n\pi b/a)} & n = 1,3,5,\dots \\ 0 & n = 2,4,6,\dots \end{cases}$$

So: $$\varphi(x,y) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4U_0}{n\pi \sinh(n\pi b/a)} \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi y}{a}\right)$$

Ex: Two semi-infinite, grounded conducting planes are parallel to each other with a separation of d. The finite end is closed by a conducting plane held at electric potential φ_0, and is isolated from the semi-infinite grounded conducting plane with a small gap. Find the **electric potential in the slot** constructed by the three conducting planes.



Solution: Select **rectangular** coordinate system. Since the conducting plane is **infinite** in the z-direction, the potential in the slot must be **independent** of z, and this is a **two-dimensional** problem. The Laplace's Equation for the electric potential becomes

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0$$

Using the method of separation of variables, and let

$$\varphi(x,\ y) = X(x)Y(y)$$

The boundary conditions for the electric potential in the slot can be expressed as

$$\varphi(x,\ 0) = 0 \qquad \varphi(x,\ d) = 0$$

$$\varphi(0,\ y) = \varphi_0 \qquad \varphi(\infty,\ y) = 0$$

In order to satisfy the boundary conditions $\varphi(x,\ 0) = 0$ and $\varphi(x,\ d) = 0$, the solution of $Y(y)$ should be selected as

$$Y(y) = A\sin k_y y + B\cos k_y y$$

From the boundary condition $\varphi(x,\ 0) = 0$, we have $\varphi = 0$ at $y = 0$, and the constant $B = 0$. In order to satisfy $\varphi(x,\ d) = 0$, the constant k_y should be

$$k_y = \frac{n\pi}{d},\ n = 1,\ 2,\ 3,\ \cdots$$

We find $$Y(y) = A\sin\left(\frac{n\pi}{d}y\right)$$

Since $k_x^2 + k_y^2 = 0$, we obtain

$$k_x = \mathrm{j}\frac{n\pi}{d}$$

The constant k_x is an imaginary number, and the solution of $X(x)$ should be

$$X(x) = C\mathrm{e}^{\frac{n\pi}{d}x} + D\mathrm{e}^{-\frac{n\pi}{d}x}$$

Since $\varphi \to \infty$ at $x = 0$, the constant $C = 0$, and

$$X(x) = D\mathrm{e}^{-\frac{n\pi}{d}x}$$

Then $$\varphi(x, y) = C\mathrm{e}^{-\frac{n\pi}{d}x}\sin\left(\frac{n\pi}{d}y\right)$$

Where the constant $C = AD$.

Since $\varphi = \varphi_0$ at $x = 0$, and we have

$$\varphi_0 = C\sin\left(\frac{n\pi}{d}y\right)$$

The right side of the above equation is variable, since **C** and n are not fixed. To satisfy the requirement at $x = 0$, one needs to take the linear combination of the equation as the solution, leading to

$$\varphi(x,y) = \sum_{n=1}^{\infty} C_n \mathrm{e}^{-\frac{n\pi}{d}x} \sin\left(\frac{n\pi}{d}y\right)$$

In order to satisfy the boundary condition $x = 0,\ \varphi = \varphi_0$,

and we have

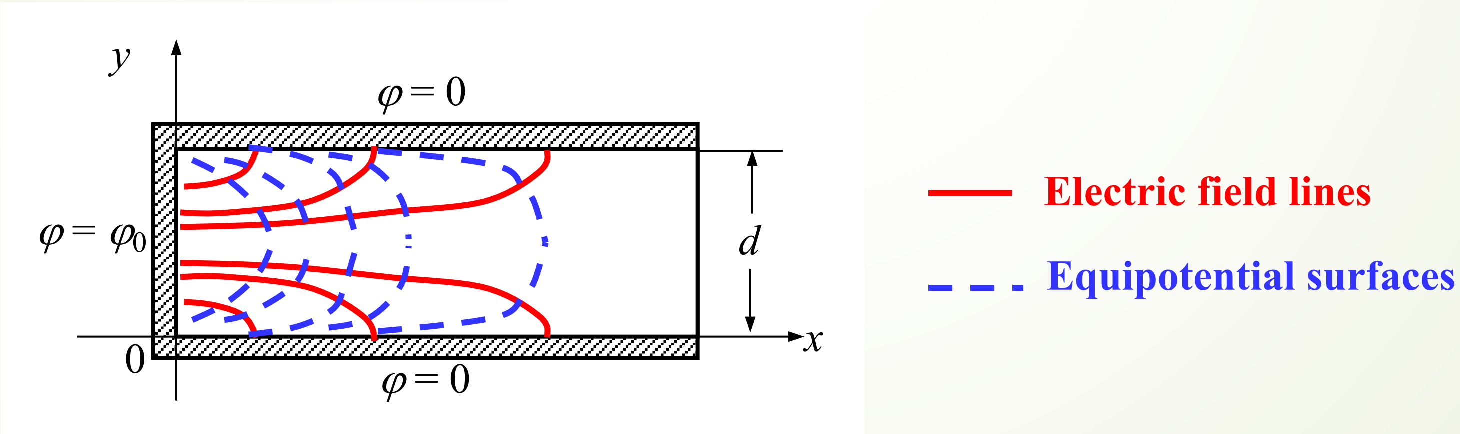
$$\varphi_0 = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{d}y\right), \quad 0 < y < d$$

The right side of the above equation is Fourier series. By using the orthogonality between the terms of a Fourier series, the coefficients C_n can be found as

$$C_n = \begin{cases} \dfrac{4\varphi_0}{n\pi}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even} \end{cases}$$

Finally, we find the electric potential in the slot as

$$\varphi(x,y) = \frac{4\varphi_0}{\pi} \sum_{n}^{\infty} \frac{1}{n} e^{-\frac{n\pi}{d}x} \sin\left(\frac{n\pi}{d} y\right) \qquad n = 1,\ 3,\ 5, \cdots$$



Summary

- **Uniqueness:** a solution of an electrostatic problem satisfying its boundary conditions is the only possible solution, irrespective of the method by which the solution is obtained.

- **Method of Images:** when the conducting bodies have boundaries of a simple geometry, the method of images may be used.

- **Boundary-Value Problems:** three types, correctly describe;

- **Method of separation of variables:** transform the high dimensional problem into one dimensional problem

Homework

4-1, 4-6, 4-7, 4-15, 4-18,
4-21